Week 1 : Linear Algebra

A, B, C	capital letters represent matrices
u, v, w	lowercase letters represent vectors
A of size $m \times n$ or $(m \times n)$	matrix A has m rows and n columns
A^T	the transpose of matrix A
v^T	the transpose of vector v
A^{-1}	the inverse of matrix A
$\det(A)$	the determinant of matrix A
AB	matrix multiplication of matrices A and B
$u \cdot v; \langle u, v \rangle$	dot product of vectors u and v
$\mathbb{R}$	the set of real numbers, e.g. $0, -0.642, 2, 3.456$
$\mathbb{R}^2$	the set of two-dimensional vectors, e.g. $v = \begin{bmatrix} 1 & 3 \end{bmatrix}^T$
$\mathbb{R}^n$	the set of n-dimensional vectors
$v \in \mathbb{R}^2$	vector v is an element of $\mathbb{R}^2$
$\lvert v \rvert_1$	L1-norm of a vector
$\lvert v \rvert_2; \lvert v \rvert; \lVert v \rVert$	L2-norm of a vector
$T : \mathbb{R}^2 \to \mathbb{R}^3; T(v) = w$	transformation T of a vector $v \in \mathbb{R}^2$ into the vector $w \in \mathbb{R}^3$

Week 1: Linear Algebra – Systems of Linear Equations

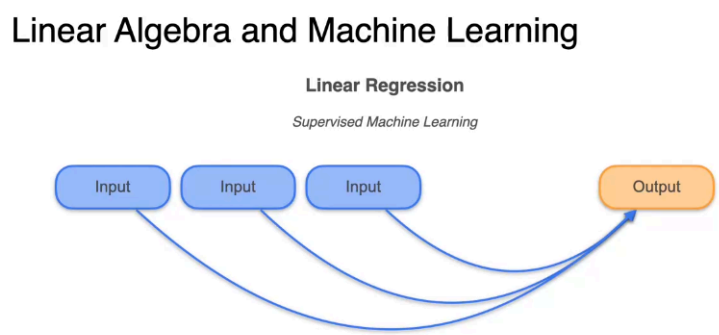
I. Linear Algebra Applied I

Introduction

This week, you will learn what a **system of linear equations** is and how it can be represented. Systems can be visualized as **lines in a plane** or as **matrices**, which are arrays of numbers. You will also explore **singular and non-singular systems**, which are important concepts in linear algebra. Linear algebra is widely used in many fields, especially in **machine learning**, where it provides the mathematical foundation for many algorithms.

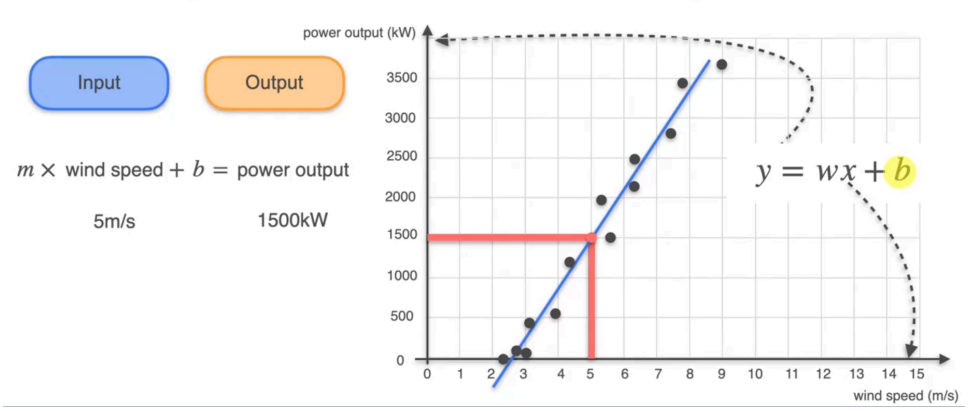
Math and Machine Learning

While this course focuses on mathematics, you will see **machine learning examples** to help understand how linear algebra is applied in practice. Even if you don't fully understand all machine learning details, the goal is to **build a strong mathematical foundation** to support future studies in machine learning.



Linear Regression with One Feature

A key example is **linear regression**, a supervised learning method. For instance, predicting the **power output of a wind turbine** using **wind speed** as a single feature can be modeled with the equation y=wx+by = wx + by=wx+b, where www is the weight and bbb is the bias. The goal is to find the **best weight and bias** to fit the data points as closely as possible.



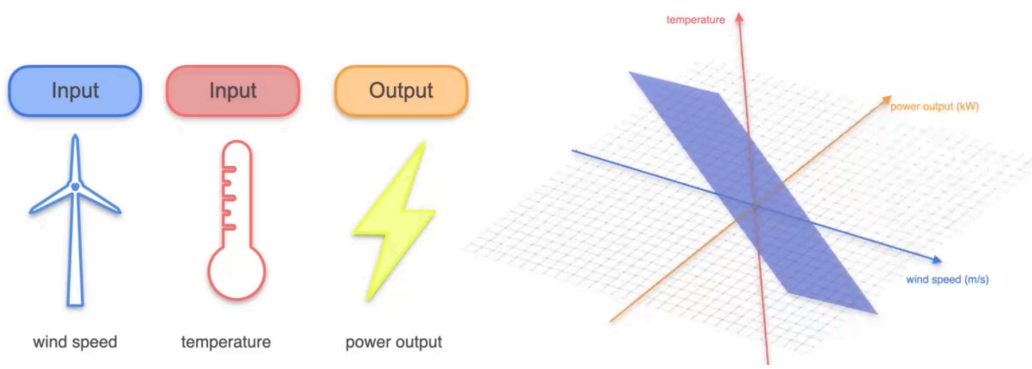
Linear Regression with Multiple Features

When additional features are included, such as **temperature, pressure, or humidity**, the equation expands:

$$w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b = y$$

TARGET

Each feature has its own weight, and the model can be visualized as a **plane or hyperplane** in higher-dimensional space. The idea is to **adjust weights and bias** to make accurate predictions.



Representing Datasets as Systems of Equations

Each record in a dataset produces one equation. With many records, you get a **system of linear equations**, which is a fundamental concept in linear algebra. Solving this system means finding the weights and bias that satisfy all equations as closely as possible.

$$w_1 x_1^{(1)} + w_2 x_2^{(1)} + \dots + w_n x_n^{(1)} + b = y^{(1)}$$

$$w_1 x_1^{(2)} + w_2 x_2^{(2)} + \dots + w_n x_n^{(2)} + b = y^{(2)}$$

$$w_1 x_1^{(3)} + w_2 x_2^{(3)} + \dots + w_n x_n^{(3)} + b = y^{(3)}$$

$$\vdots$$

$$w_1 x_1^{(m)} + w_2 x_2^{(m)} + \dots + w_n x_n^{(m)} + b = y^{(m)}$$

subscripts except that we

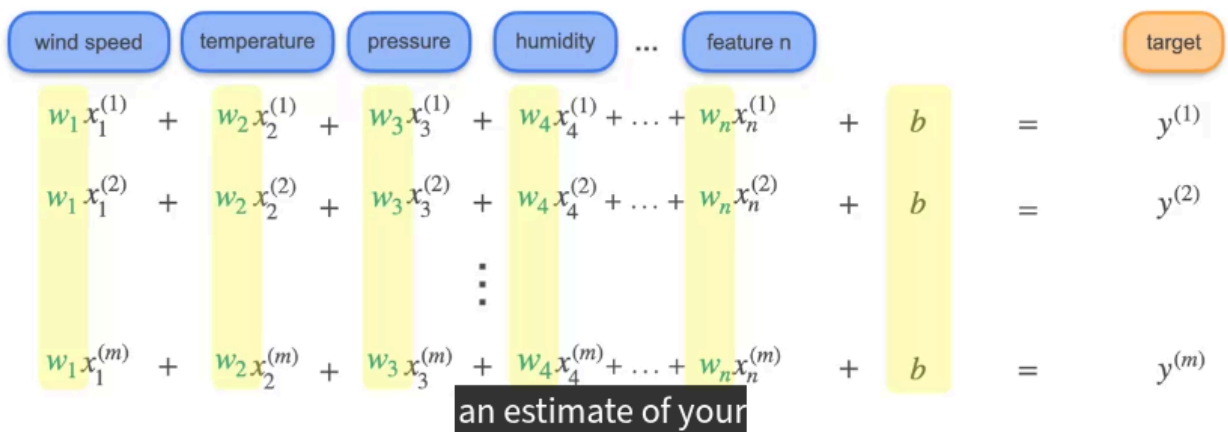
Summary

Linear regression demonstrates how **linear algebra underpins machine learning**. Systems of linear equations, matrices, weights, and biases are key tools that allow us to model and solve real-world problems. This week, you'll study these concepts in depth and see how they apply to datasets and machine learning models.

II. Linear Algebra Applied II

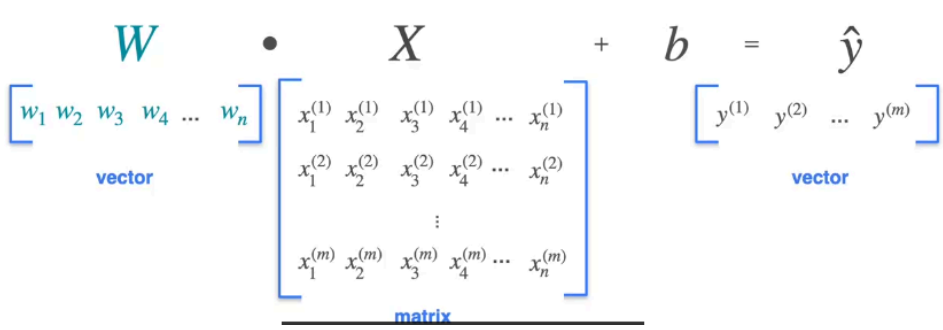
In the previous video, we looked at a **linear regression scenario** predicting the electrical power output from a wind turbine. The dataset contained multiple features, such as wind speed, temperature, atmospheric pressure, and humidity, labeled as x1,x2,…,xn.

Each row of data was denoted with a superscript to identify which record it belonged to. The **weights** (w1,w2,…,wn) multiplied their corresponding features, and a **bias** term bbb was added to estimate the target y, the power output. The xs and ys are unique for each row, but the weights and bias remain the same across all rows.



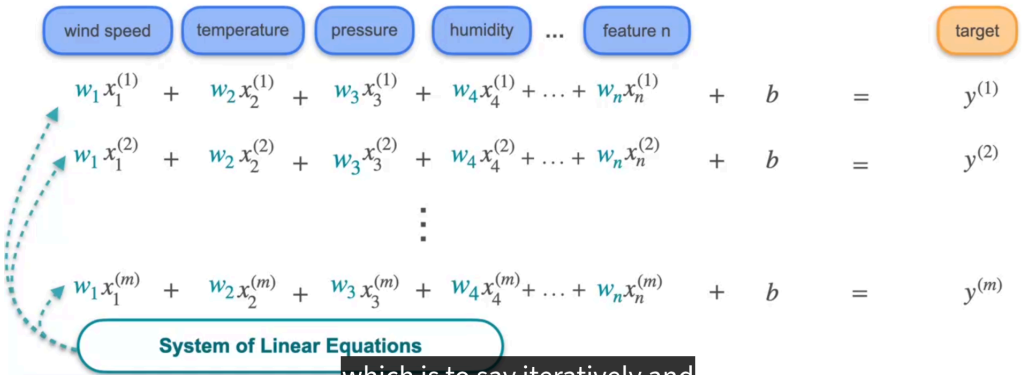
Vector and Matrix Representation

Instead of writing each equation separately, we can represent the system using **vectors and matrices**. The weights form a vector w, the rows of features form a matrix X, and the bias is added to compute a vector of target values y. This compact representation simplifies the linear model and shows how **linear algebra underpins many machine learning techniques**. Even if you are not familiar with vectors or matrices, you can think of them as **lists or grids of numbers** that can be manipulated to perform calculations efficiently.



Solving Systems of Equations

A linear regression model is essentially a **system of linear equations**. If a perfect set of weights and bias existed, you could solve for them analytically using basic algebra. In practice, linear regression solves this system **empirically**, finding the best-fit solution iteratively and approximately. This illustrates the practical use of linear algebra in machine learning.



Learning Goals for the Week

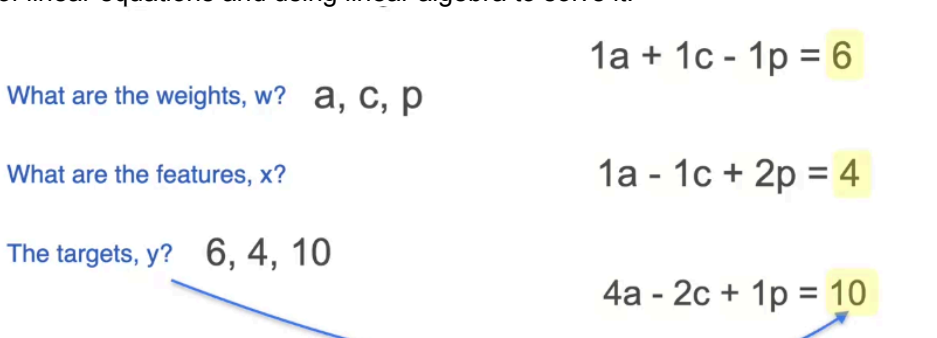
This week, you will start with **basic vector and matrix operations** and gradually explore **systems of linear equations**, their representations, and how to compute properties like **determinants** and **singularity**. You will learn to manipulate these systems step by step, starting from simple sentences that form equations, moving to matrices, and eventually solving the systems.

Example: Student Scores

To illustrate, consider a hypothetical example with scores in **Linear Algebra, Calculus, and Probability & Statistics**. You are given relationships between scores in three equations. These can be written as a **system of linear equations** with the scores as variables. You can then ask questions such as:

- Can this system be solved?
- Is it singular or non-singular?
- How can it be represented as a **matrix and vector**?
- Can you calculate its **determinant**?

These questions mirror the process used in linear regression: representing a real-world problem as a system of linear equations and using linear algebra to solve it.



What is REALLY happening (truth table)

In reality (linear algebra)

- **a, c, p** → unknown variables
- **1, 2, 4 (and −1, −2)** are called **coefficients.** More precisely: **entries of the coefficient matrix**
- The equations are **constraints**
- Goal: **solve exactly**
- This is a **system of linear equations**

In the analogy (machine learning language)

- **a, c, p** → called “weights”
- **1, 2, 4** are called **features**
- Coefficients → called “features”
- Right-hand side → called “targets”
- Goal: **show structural similarity**

They are **borrowing ML vocabulary**, not changing the math.

III. System of sentences

In linear algebra, the first topic is **systems of linear equations**, but before tackling equations, it’s crucial to understand the **language of mathematics**. Equations are like sentences that convey information about the world. Just as sentences can combine to give more meaning, equations in a system combine to provide more information.

To illustrate, consider **systems of simple sentences** about one dog and one cat, each with a single color:

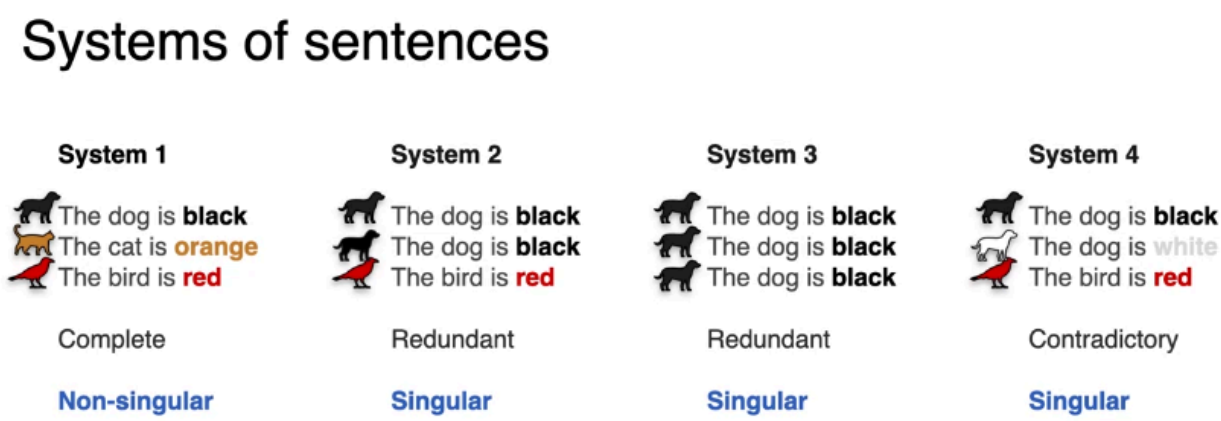
1. **System 1:** “The dog is black” and “The cat is orange.”
    - Each sentence gives unique information.
    - This system is **complete** (non-singular) because the number of sentences equals the number of distinct pieces of information.

2. **System 2:** “The dog is black” and “The dog is black.”
    - The sentences repeat information.
    - This system is **redundant**, carrying only one piece of information, and is considered **singular**.

3. **System 3:** “The dog is black” and “The dog is white.”
    - The sentences contradict each other.
    - This system is **contradictory** and also **singular**.

Key takeaway: A system's usefulness depends on how much information it conveys.

- **Non-singular (complete) systems:** provide maximum distinct information.
- **Singular systems:** either redundant or contradictory, offering less or conflicting information.



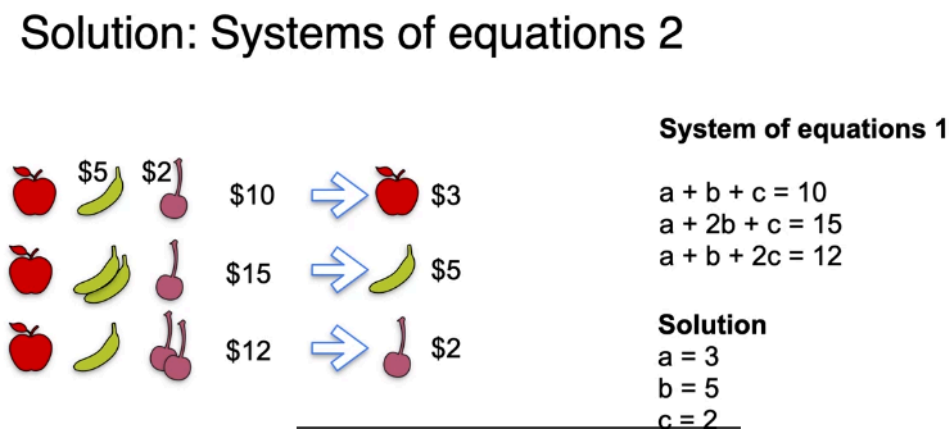
IV. System of linear equations

Equations are like sentences—they convey information. In linear algebra, a **linear equation** expresses numerical information, and a **system of linear equations** combines multiple equations to extract as much information as possible. Like sentence systems, equation systems can be **non-singular (complete)** or **singular (redundant or contradictory)**.

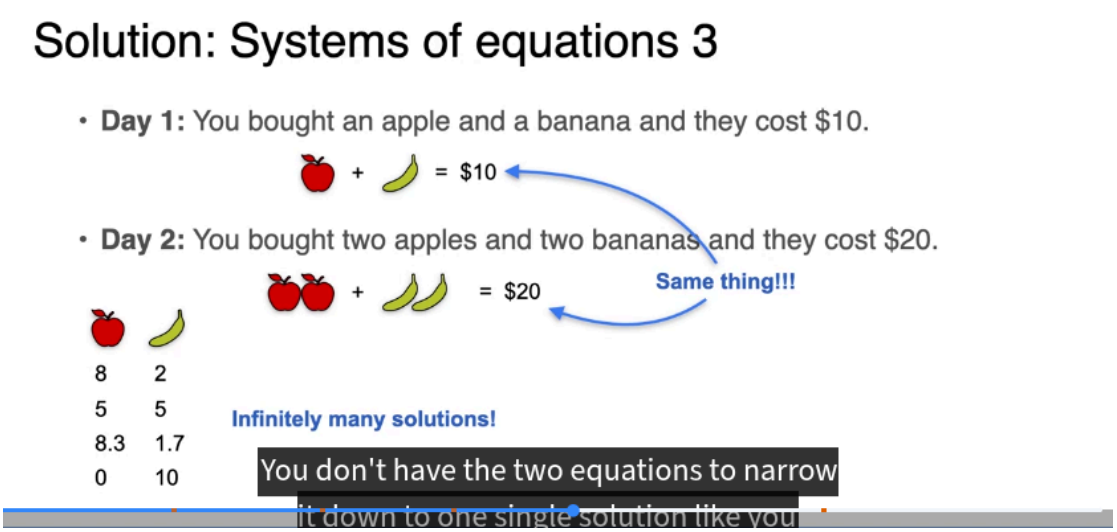
For example, consider finding fruit prices in a store where only total costs are given:

- **Two fruits, two equations:**
    - Day 1: apple + banana = $10
    - Day 2: apple + 2 bananas = $12
    - By comparing the days, the extra banana costs $2, so the apple costs $8.
    - This is a **complete system**, giving a unique solution.

- **Three fruits, three equations:**
    - Day 1: apple + banana + cherry = $10
    - Day 2: apple + 2 bananas + cherry = $15
    - Day 3: apple + banana + 2 cherries = $12
    - By comparing days, extra banana = $5, extra cherry = $2, leaving apple = $3.
    - System solved uniquely.



- **Redundant system:**
    - Day 1: apple + banana = $10
    - Day 2: 2 apples + 2 bananas = $20
    - The second equation is just twice the first—no new information.
    - The system has **infinitely many solutions**, e.g., apple = 8 & banana = 2, apple = 5 & banana = 5, etc.



In this final scenario, you try to find the prices of an apple and a banana:

- Day 1: apple + banana = $10
- Day 2: 2 apples + 2 bananas = $24

If there were no mistakes, twice the first day's total should equal the second day's total, i.e., $20. But the register charged $24—$4 extra. Since there are no discounts or extra fees, this indicates an inconsistency.

Just like the contradictory sentence "the dog is black" and "the dog is white," these equations **contradict each other**. Therefore, the system has **no solution**.

Solution: Systems of equations 4

- **Day 1:** You bought an apple and a banana and they cost $10.

- **Day 2:** You bought two apples and two bananas and they cost $24.

Contradiction!

No solutions!

Solutions: More systems of equations

System 1	System 2	System 3
$a + b + c = 10$ $a + b + 2c = 15$ $a + b + 3c = 20$	$a + b + c = 10$ $a + b + 2c = 15$ $a + b + 3c = 18$	$a + b + c = 10$ $2a + 2b + 2c = 20$ $3a + 3b + 3c = 30$
Infinitely many sols. $c = 5$ $a + b = 5$ (0,5,5), (1,4,5), (2,3,5), …	**No solutions** From 1st and 2nd: $c = 5$ From 2nd and 3rd: $c = 3$	**Infinitely many solutions** Any 3 numbers that add to 10 work. (0,0,10), (2,7,1), …

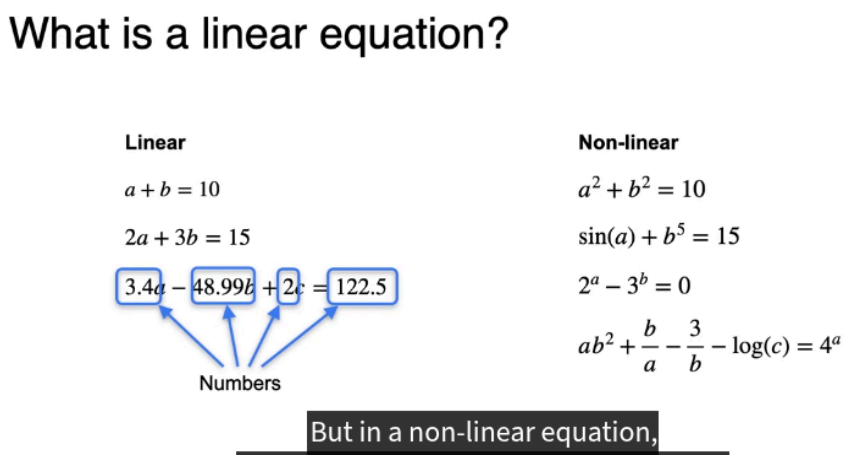
A **linear equation** is an equation where variables appear only multiplied by constants (scalars) and added or subtracted, with an optional standalone number. Examples include:

- Simple: a+b=10
- Slightly complex: 2a+3b=15

Linear equations can have **any number of variables**, but the key rule is no powers, products of variables, or complicated functions—just scalars times variables plus constants.

In contrast, **non-linear equations** can include: squares (a2a^2a2), higher powers (b5b^5b5), products of variables (ababab), divisions (b/ab/ab/a), exponentials (2a2^a2a), trigonometric or logarithmic functions, etc.

Linear algebra focuses solely on **linear equations**, because they are simpler and allow systematic manipulation and extraction of information. Non-linear equations are more complex and not the main focus here.



V. System of equations as lines and planes

V.1 Two variables

Linear equations can be **visualized geometrically**. With **two variables**, each equation corresponds to a **line** in the coordinate plane: one axis for each variable. With three variables, equations form **planes** in space, and more variables create higher-dimensional analogs.

For example, the equation a+b=10a + b = 10a+b=10 (apple + banana = 10) can be plotted:

- Points like (10,0), (0,10), (4,6), and (8,2) satisfy the equation.
- Even negative pairs like (-4,14) or (12,-2) satisfy it mathematically, though they may not make sense in real life.
- All these points form a **line**, meaning every point on the line is a solution.

Another equation, a+2b=12a + 2b = 12a+2b=12, also forms a line:

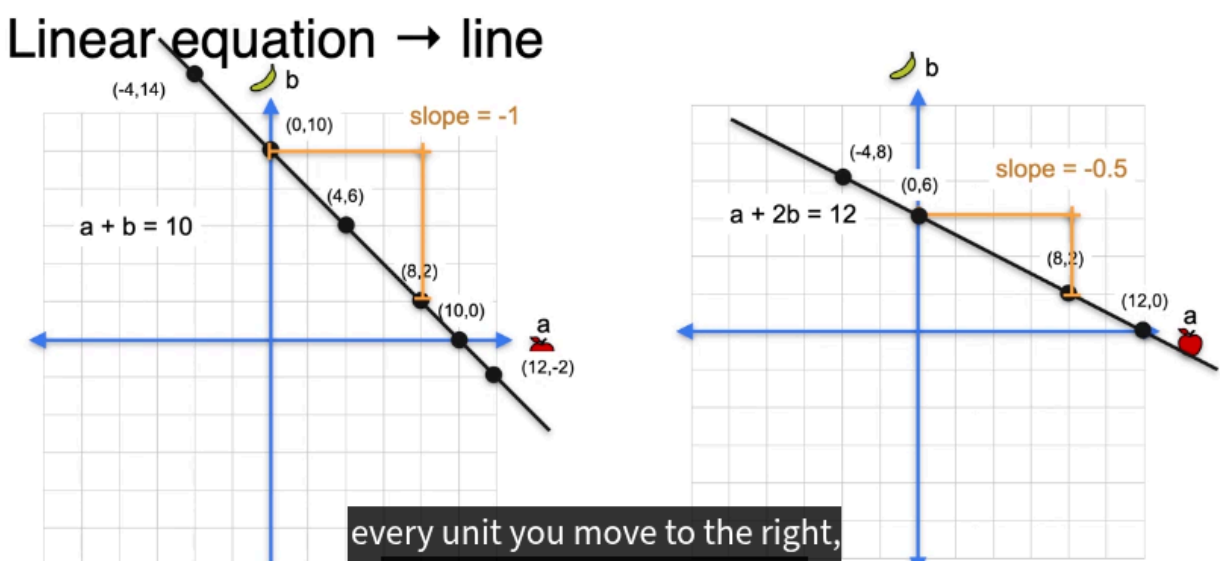
- Solutions include (0,6), (12,0), (8,2), or (-4,8).
- Each point on this line satisfies the equation.

Lines can be described using **slope** and **y-intercept**:

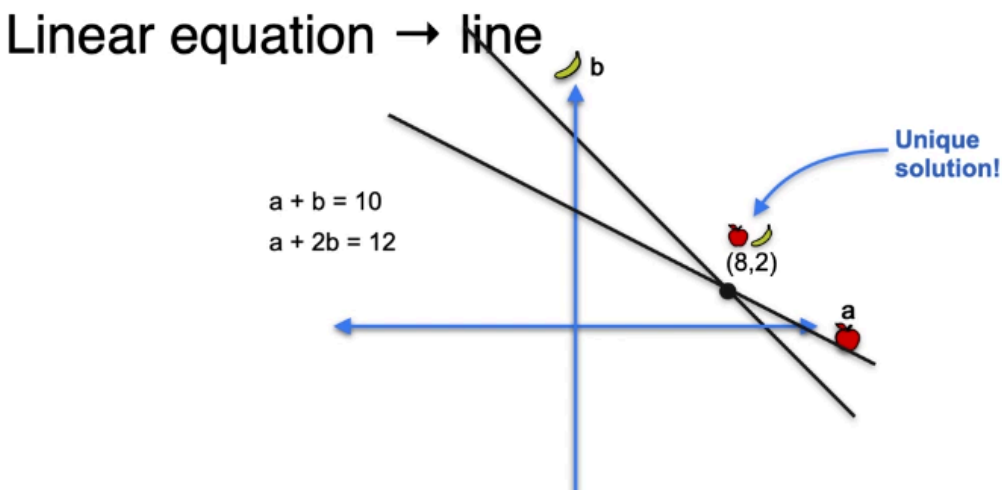
- For a+b=10, slope = -1 (line goes down 1 unit for every 1 unit right), y-intercept = 10.
- For a+2b=12 slope = -0.5, y-intercept = 6.

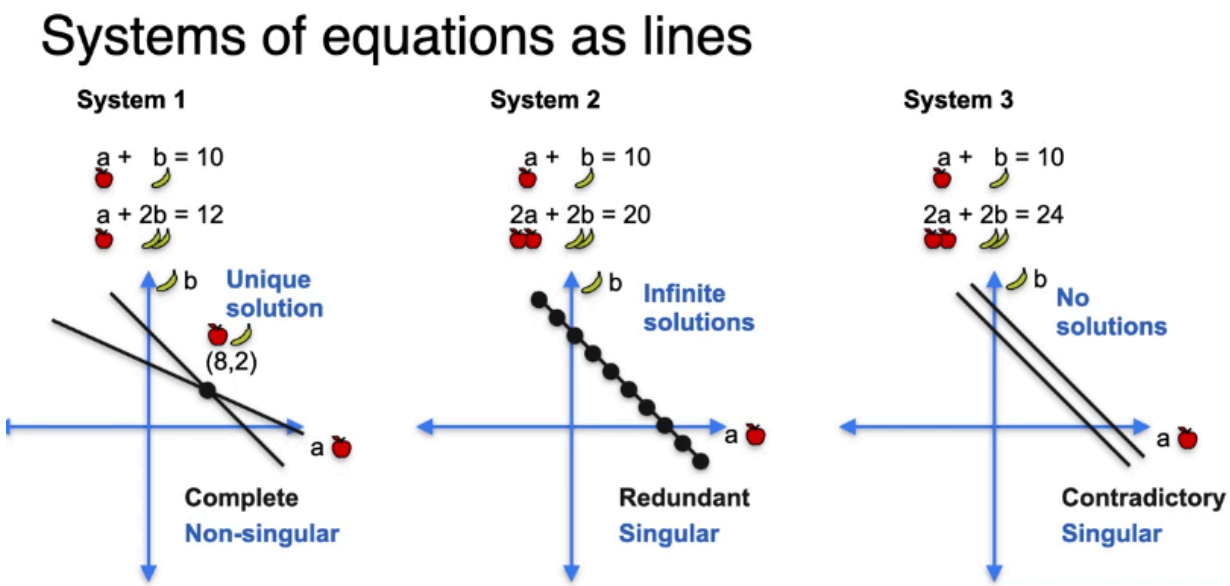
By visualizing **systems of equations as multiple lines**, you can see:

- **Non-singular system:** lines intersect at a single point (unique solution).
- **Redundant system:** lines overlap (infinitely many solutions).
- **Contradictory system:** lines are parallel and never meet (no solution).



The two lines intersect at a single point, (8,2), meaning a = 8 and b = 2. This is the **unique solution** to the system—matching the algebraic solution, now seen **geometrically**.





V.2 Three variables

A **linear equation with three variables** corresponds to a **plane** in three-dimensional space. The axes represent aaa (horizontal), bbb (vertical), and ccc (coming out of the screen).

For example, the equation a+b+c=1 includes points like:

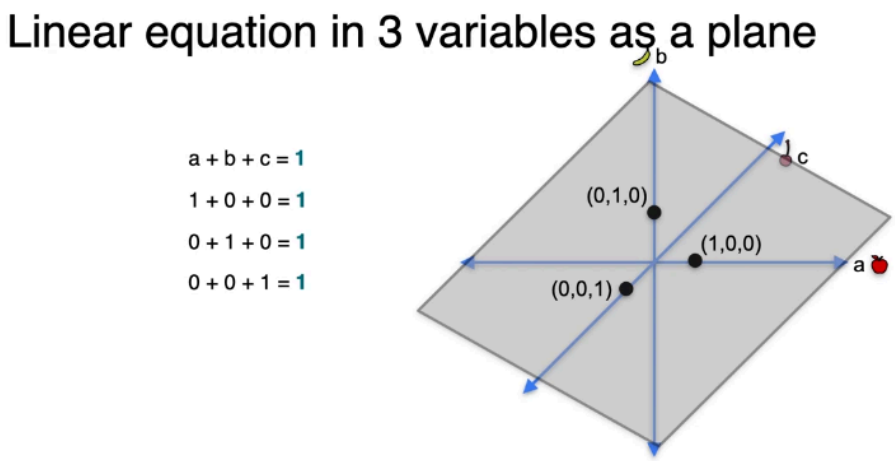
- (1,0,0) → 1+0+0=1
- (0,1,0) → 0+1+0=1
- (0,0,1) → 0+0+1=1

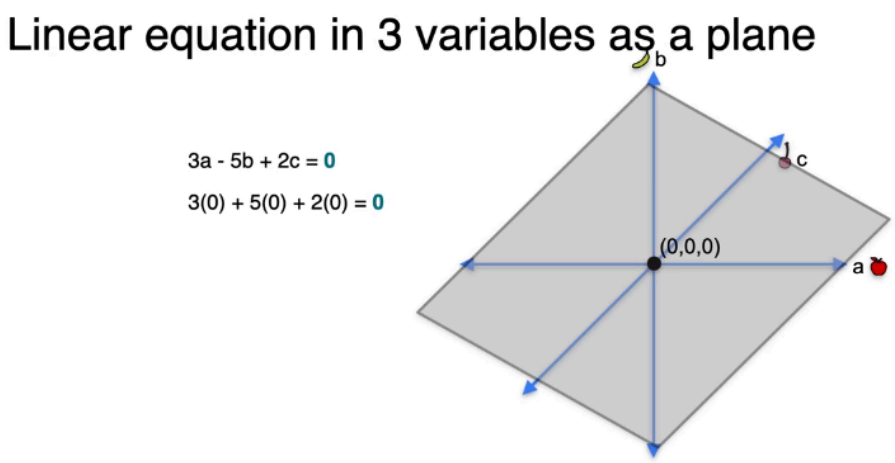
These three points define a **plane**, and every point on this plane is a solution to the equation.

Similarly, for equations like 3a−5b+2c=0, the plane passes through the **origin** (0,0,0), because setting all variables to zero satisfies the equation.

In short:

- **Two variables** → **line** in 2D
- **Three variables** → **plane** in 3D
- The plane contains **all solutions** of the equation.

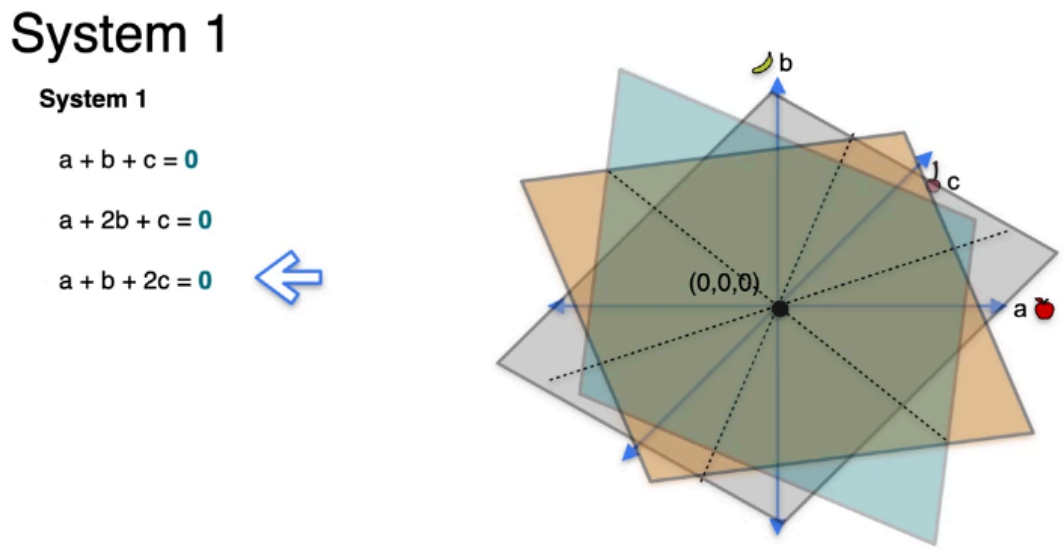




Example 1 – Non-singular system:
Equations:

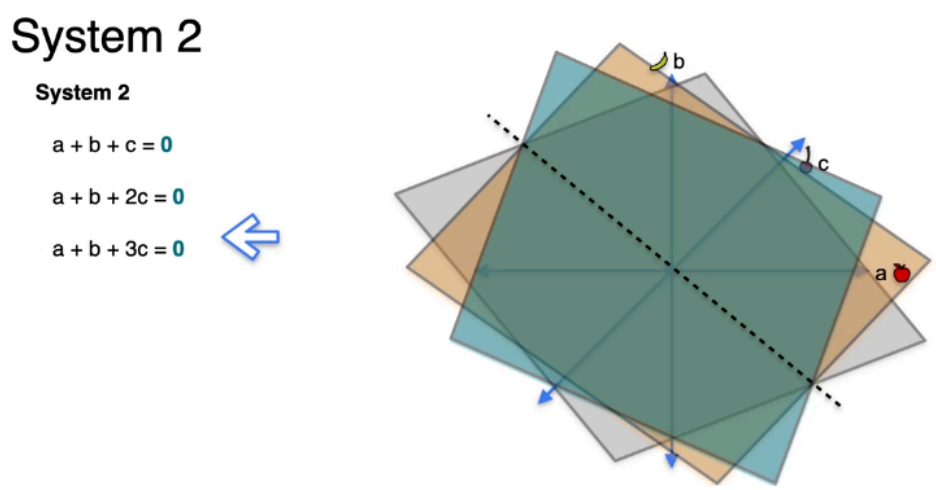
- a+b+c=0
- a+2b+c=0
- a+b+2c=0
- Each equation represents a plane through the origin.

- The intersection of all three planes is a **single point**: (0,0,0).
- This system has a **unique solution** → **non-singular**.



Example 2 – Singular system (line of solutions):

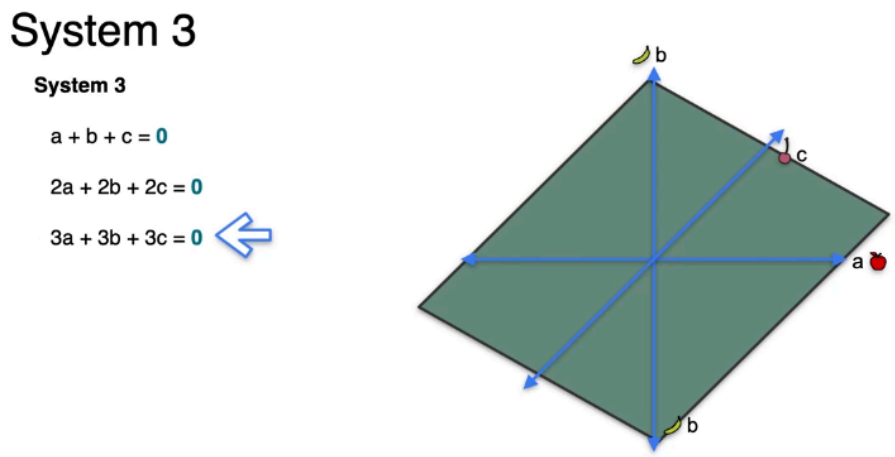
- First two planes intersect along a **line** through the origin.
- The third plane also passes through that line.
- The intersection is now an **entire line**, not a single point.
- The system has **infinitely many solutions** → **singular**.



Example 3 – Singular system (entire plane):
Equations:

- a+b+c=0
- 2a+2b+2c=02
- 3a+3b+3c=03
- All three planes are **the same plane**.
- Every point on this plane is a solution.
- The system has **infinitely many solutions** → **singular**.

Redandunt



In short:

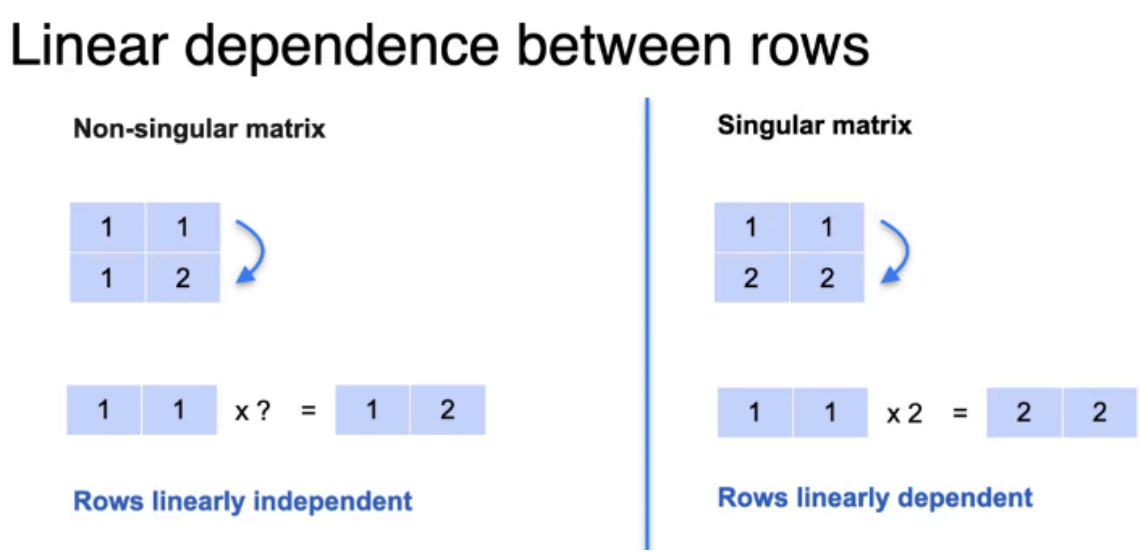
- **Unique point intersection → non-singular**
- **Line or plane intersection → singular**

https://www.coursera.org/learn/machine-learning-linear-algebra/supplement/ZR44d/interactive-tool-system-of-equations-as-planes-3x3

VI. Singular vs Non Singular Matrix

Imagine we have 2*2 matrix :

- If row2 = row1 * n -> Rows linearly dependent
- If row2 = row1 * ? -> Rows linearly independent

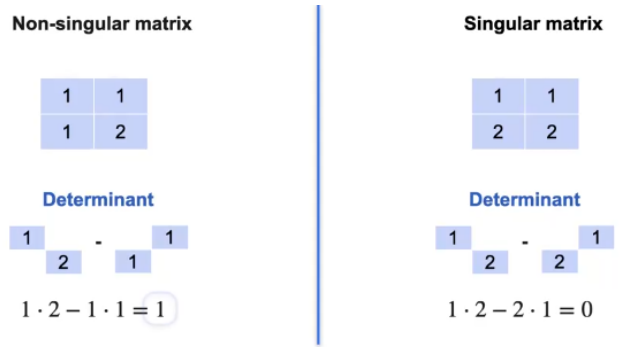


Matrix is singular when determinant = 0

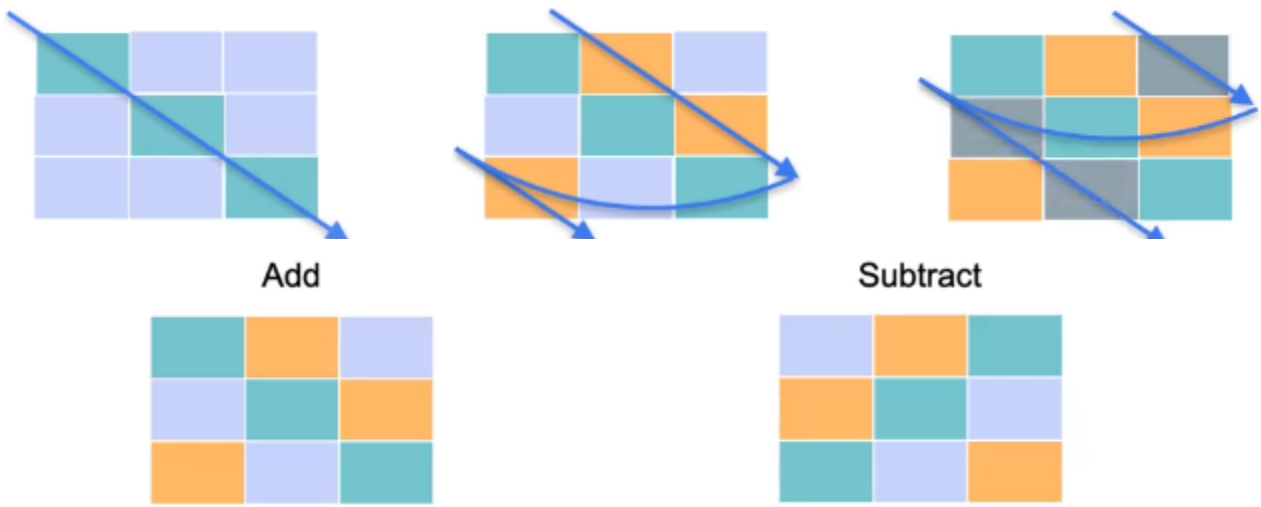
For 2*2 matrix

a	b
c	d

determinant = a*b - b*c



For 3*3 matrix or n*n matrix



Special case :

1	1	1
0	2	2
0	0	3

If all values under or above diagonal line
The determinant is the product of the main diagonal (either singular or non singular)

#==========

VII. Linear dependence

Dependency = *Can one row in the matrix be obtained as a result of operations on the other rows?*

System of sentences is singular if the second sentence carries the same information as the first one. This is the concept of liner dependent.

singularity express dependency.

Singularity expresses a loss of information.

- Some equations are **redundant** (they repeat the same information), **or**
- Some equations **contradict each other**

Because of this, the system **cannot give a unique solution**.

In terms of linear systems

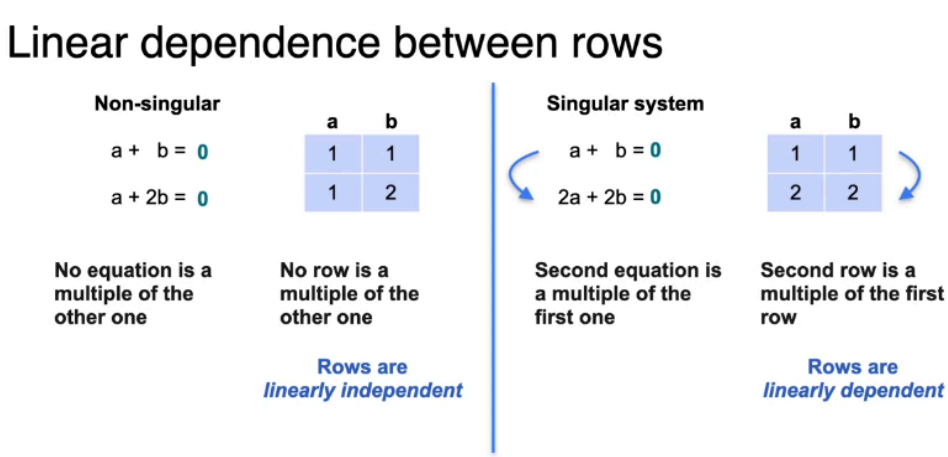
If a matrix is **singular**, then the corresponding system of linear equations:

- has **no solution or**
- has **infinitely solutions**

So singularity expresses that the system is **not fully independent**. If AAA is non-singular, there is **exactly one unique solution**.

Geometric meaning (very important intuition)

- A **non-singular** matrix transforms space without collapsing it
- A **singular** matrix squashes space into a lower dimension
    - a plane → a line
    - a line → a point



Linear dependence and independence

$a = 1$
$b = 2$
$a + b = 3$

$$\begin{array}{rl} & a + 0b + 0c = 1 \\ + & 0a + b + 0c = 2 \\ \hline & a + b + 0c = 3 \end{array}$$

1	0	0
0	1	0
1	1	0

Row 1 + Row 2 = Row 3

Row 3 **depends** on rows 1 and 2

Rows are **linearly dependent**

Linear dependence and independence

$a + b + c = 0$
$a + b + 2c = 0$
$a + b + 3c = 0$

$$\begin{array}{rl} & a + b + c = 0 \\ + & a + b + 3c = 0 \\ \hline & 2a + 2b + 4c = 0 \end{array}$$

$\div 2$

$a + b + 2c = 0$

1	1	1
1	1	2
1	1	3

Average of Row 1 and Row 3 is Row 2
Row 2 **depends** on rows 1 and 3

Linear dependence and independence

$a + b + c = 0$
$a + 2b + c = 0$ → No relations between equations
$a + b + 2c = 0$

1	1	1
1	2	1
1	1	2

No relations between rows

Rows are **linearly independent**

Solution: Linear dependence and independence

Problem: Determine if the following matrices have linear dependent or independent rows

1	0	1
0	1	0
3	2	3

3Row1 + 2Row2 = Row3

Dependent
(singular)

1	1	1
1	1	2
0	0	-1

Row1 - Row2 = Row3

Dependent
(singular)

1	1	1
0	2	2
0	0	3

No relations

Independent
(Non-singular)

1	2	5
0	3	-2
2	4	10

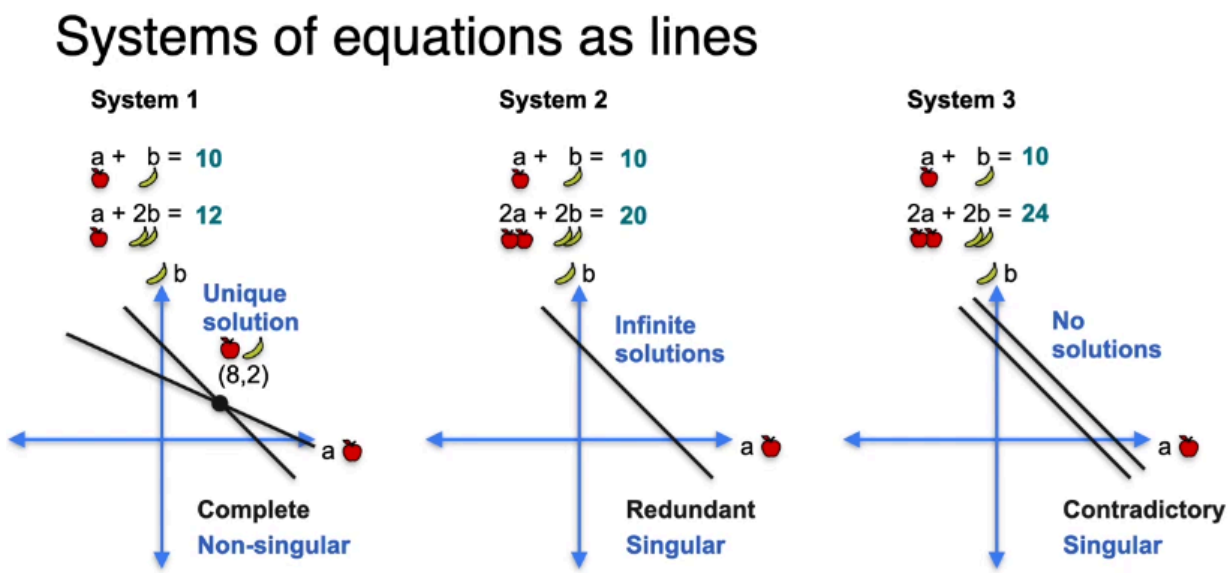
2Row1 = Row3

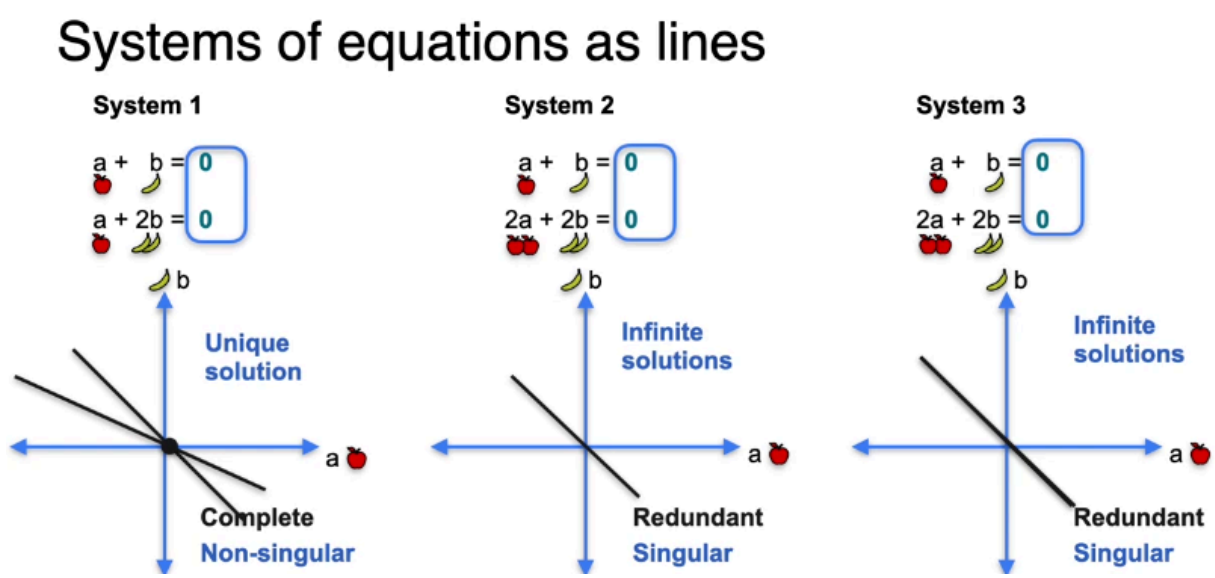
Dependent
(singular)

#==========

Geometric Notion of singularity

We have 3 different system of equations:





The constant **(10, 12, 20, 24)** is **not important** when we want to know whether a system is **singular or non-singular**, which is what we mainly care about in this course. Concepts such as **completeness** and **redundancy** are **much less important** for our purposes.

One-sentence takeaway (remember this)

👉 **A machine learning model is a large, noisy system of sentences that cannot be solved exactly, so we use linear algebra to find the best approximation.**